TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

# Thermal Conductivity of Metals

*Searle's bar method: hold a copper bar between steam and a stream of cold water, read the temperature gradient along it, and count the heat the water carries away — that heat is what the bar conducted.*

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 15 · LEVEL 1 TERM 2

## 1. Aim and theory

**Aim:** to determine the thermal conductivity k of a metal (copper) by Searle's bar method. In the steady state, the heat conducted per second along a bar of cross-section A whose temperature falls from $\theta_1$ to $\theta_2$ over a length d is

$$Q/t = k\,A\,(\theta_1 - \theta_2)\,/\,d$$

The same heat is removed by the cooling water flowing round the cold end. If water of mass m is collected in time t and warms from $\theta_3$ (inlet) to $\theta_4$ (outlet), it carries away

$$Q/t = (m/t)\,s\,(\theta_4 - \theta_3)\,, \quad s = 4186\ \mathrm{J\ kg^{-1}\ K^{-1}}$$

In the steady state the two rates are equal — nothing accumulates in the bar — so

$$k = m\,s\,(\theta_4 - \theta_3)\,d\,/\,(\,A\,(\theta_1 - \theta_2)\,t\,)$$

$\theta_1$ and $\theta_2$ are read by thermometers set into holes drilled in the bar a distance d apart. The gradient $(\theta_1 - \theta_2)/d$ must be measured on the straight, lagged part of the bar, clear of both ends, where the temperature falls uniformly.

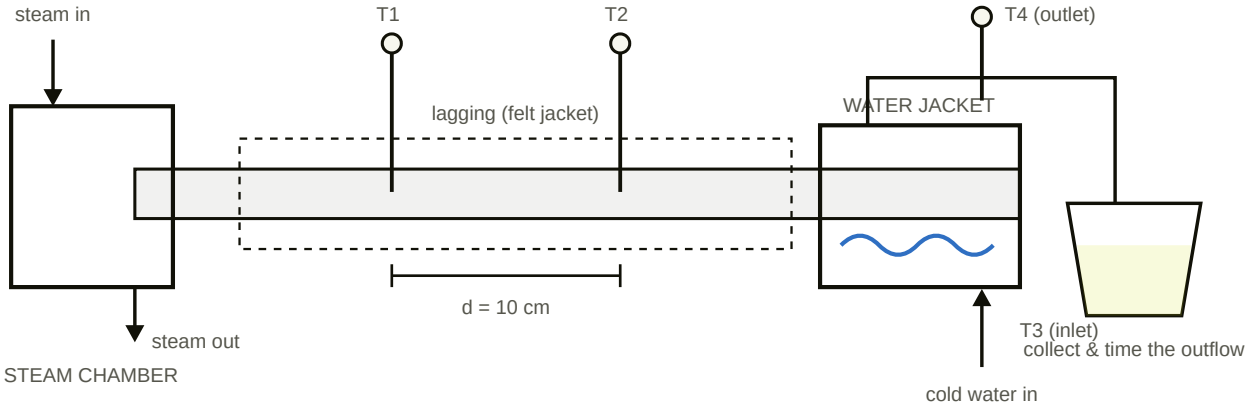



Figure 1 — Searle's apparatus. Steam holds the hot end steady; T1 and T2 read the temperature gradient along the lagged bar; the cooling water (in at T3, out at T4) carries away exactly the heat the bar conducts.

## 2. Apparatus and procedure

- **Apparatus:** Searle's bar (copper rod in a lagged wooden box) with steam chamber and water jacket, four thermometers (0.5 °C or finer), steam generator, constant-head water supply, measuring cylinder, stopwatch, vernier calipers, metre scale.
- Measure the bar's diameter with calipers at several places and find $A = \pi D^2/4$. Measure d, the distance between the T1 and T2 holes, between hole centres.
- Set the thermometers in their holes with a drop of mercury or oil in each hole for good thermal contact. Start the steam and a slow, steady stream of cooling water from the constant-head tank.
- Wait for the steady state — typically 25–30 minutes — until all four thermometers have stood constant for at least 5 minutes. Read $\theta_1$, $\theta_2$, $\theta_3$, $\theta_4$.
- Without disturbing the flow, collect the outflow in a measuring cylinder for a measured time t (about 2 minutes) and weigh or measure the mass m of water collected.
- Repeat with a different water flow rate; the value of k should agree, because a faster flow lowers $\theta_4 - \theta_3$ but raises m/t in proportion.

## 3. Worked calculation

Steady readings: bar diameter D = 4.0 cm, so $A = \pi(0.020\ \text{m})^2 = 1.257 \times 10^{-3}\ \text{m}^2$; d = 0.100 m; $\theta_1$ = 84.0 °C, $\theta_2$ = 64.0 °C. Water collected: m = 240 g in t = 120 s (flow $2.0 \times 10^{-3}$ kg $s^{-1}$); inlet $\theta_3$ = 26.0 °C, outlet $\theta_4$ = 38.0 °C.

Heat carried away per second:

$$Q/t = 2.0 \times 10^{-3} \times 4186 \times (38.0 - 26.0) = 100.5\ \text{W}$$

$$k = 100.5 \times 0.100 / (1.257 \times 10^{-3} \times 20.0) \approx 400 \text{ W m}^{-1} \text{ K}^{-1}$$

**RESULT**

k(copper) ≈ $4.0 \times 10^2$ W $m^{-1}$ $K^{-1}$ — against the accepted value of about 401 W $m^{-1}$ $K^{-1}$ for pure copper. A noticeably smaller result usually means the steady state was not reached, or heat escaped from the bar's surface despite the lagging (that heat is conducted but never reaches the water, so the water accounts for too little).

## 4. Precautions

- Do not rush the steady state: readings taken while $\theta_1$ and $\theta_2$ are still creeping mix stored heat with conducted heat and spoil the balance.
- Keep the water flow slow and perfectly steady — $\theta_4 - \theta_3$ of about 10 °C is a good compromise: large enough to read accurately, small enough that the jacket stays near room temperature and loses little heat of its own.
- Thermometer bulbs must make good contact in their holes (mercury or oil filling); an air gap reads low and unevenly.
- Guard the bar from draughts and keep the lagging complete — surface loss is heat the water never sees.

## Practice (viva preparation)

1. Why must the experiment wait for the steady state before any readings are taken?
2. With the worked example's gradient, what is the temperature 5 cm beyond T2 (away from the steam end)?
3. An aluminium bar of the same dimensions replaces the copper one (k ≈ 237 W $m^{-1}$ $K^{-1}$). With the same steam and the same $\theta_1 - \theta_2$, how does the required water flow compare?
4. Why is Searle's bar method suitable for good conductors but not for a poor conductor such as wood?
5. A student collects the water *before* the steady state is reached. In which direction is k in error?

### Answer key

1. Only in the steady state is the heat entering any section of the bar per second exactly equal to the heat leaving it, so the heat the water removes equals the heat conducted past T1–T2. Before that, part of the conducted heat is still being stored in the bar as its temperature creeps up.
2. The gradient is (84.0 − 64.0)/0.100 = 200 K $m^{-1}$, i.e. 10 K per 5 cm. Beyond T2 the temperature continues to fall: 64.0 − 10 = 54 °C.

3. Heat conducted per second falls in the ratio 237/400 ≈ 0.59, so for the same water temperature rise the flow must be reduced to about 0.59 of the copper value (about 1.2 g $s^{-1}$ instead of 2.0 g $s^{-1}$).
4. A good conductor gives a small, easily measured gradient over a convenient length while passing a measurable heat current. A poor conductor would need an enormous gradient (or pass an immeasurably small heat current) and surface losses would swamp the conducted heat — poor conductors are measured by the disc (Lees) method instead.
5. Too small. Before the steady state some of the conducted heat is still being stored in the bar and the cold end is still warming, so the water carries away less than the bar is conducting at T1–T2; the water's account underestimates k.

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